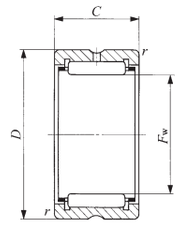
MACHINED TYPE NEEDLE ROLLER BEARINGS

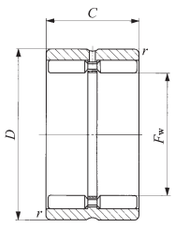
Without Inner Ring



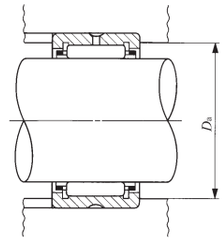




RNA49 TAF TR
RNA69 ($F_w \leqq 35$)



GTR



Shaft dia. 16 — 22mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
16	**RNA 4901**	—	—	—	—	—	18.1
	—	—	—	**TAF 162416**	—	—	23
	—	—	—	**TAF 162420**	—	—	29
	—	**RNA 6901**	—	—	—	—	30
17	—	—	—	**TAF 172516**	—	—	24.5
	—	—	—	**TAF 172520**	—	—	30.5
18	**RNA 49/14**	—	—	—	—	—	19.9
	—	—	—	**TAF 182616**	—	—	25.5
	—	—	—	**TAF 182620**	—	—	32
19	—	—	—	**TAF 192716**	—	—	27
	—	—	—	**TAF 192720**	—	—	34
20	**RNA 4902**	—	—	—	—	—	21.5
	—	—	—	**TAF 202816**	—	—	27.5
	—	—	—	**TAF 202820**	—	—	35.5
	—	**RNA 6902**	—	—	—	—	37
	—	—	—	—	**TR 203320**	—	59.5
	—	—	—	—	—	**GTR 203320**	69
21	—	—	—	**TAF 212916**	—	—	29
	—	—	—	**TAF 212920**	—	—	36
22	**RNA 4903**	—	—	—	—	—	23.5
	—	—	—	**TAF 223016**	—	—	30
	—	—	—	**TAF 223020**	—	—	37.5
	—	**RNA 6903**	—	—	—	—	40.5
	—	—	—	—	**TR 223425**	—	73.5
	—	—	—	—	—	**GTR 223425**	87

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
16	24	13	0.3	22	9 660	11 100	25 000
16	24	16	0.3	22	12 300	15 100	25 000
16	24	20	0.3	22	15 500	20 400	25 000
16	24	22	0.3	22	17 100	23 000	25 000
17	25	16	0.3	23	12 900	16 300	25 000
17	25	20	0.3	23	16 300	22 000	25 000
18	26	13	0.3	24	10 600	12 800	20 000
18	26	16	0.3	24	13 400	17 500	20 000
18	26	20	0.3	24	17 000	23 600	20 000
19	27	16	0.3	25	14 000	18 700	20 000
19	27	20	0.3	25	17 700	25 300	20 000
20	28	13	0.3	26	10 900	13 800	20 000
20	28	16	0.3	26	13 900	18 800	20 000
20	28	20	0.3	26	17 600	25 400	20 000
20	28	23	0.3	26	19 300	28 800	20 000
20	33	20	0.3	31	24 300	26 500	20 000
20	33	20	0.3	31	29 200	37 200	7 500
21	29	16	0.3	27	14 400	20 000	19 000
21	29	20	0.3	27	18 200	27 100	19 000
22	30	13	0.3	28	11 700	15 600	18 000
22	30	16	0.3	28	14 900	21 200	18 000
22	30	20	0.3	28	18 900	28 700	18 000
22	30	23	0.3	28	20 800	32 500	18 000
22	34	25	0.3	32	29 100	36 800	18 000
22	34	25	0.3	32	37 900	57 800	7 000

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. TAF series with a roller set bore diameter F_w of 26 mm or less have no oil hole. In others, the outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

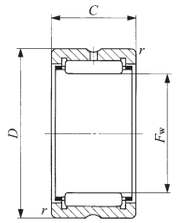
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

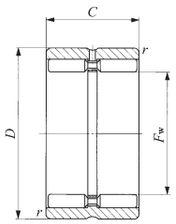
Without Inner Ring



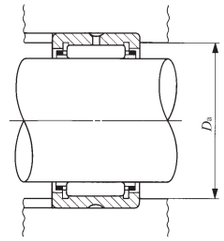




RNA49 TAF TR
RNA69 ($F_w \leqq 35$)



GTR



Shaft dia. 24 — 30mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
24	—	—	—	TAF 243216	—	—	32
	—	—	—	TAF 243220	—	—	40.5
25	—	—	—	TAF 253316	—	—	33.5
	—	—	—	TAF 253320	—	—	42
	RNA 4904	—	—	—	—	—	55.5
	—	RNA 6904	—	—	—	—	95.5
	—	—	—	—	TR 253820	—	71
	—	—	—	—	TR 253825	—	89
	—	—	—	—	—	GTR 253820	81.5
	—	—	—	—	—	GTR 253825	104
26	—	—	—	TAF 263416	—	—	34.5
	—	—	—	TAF 263420	—	—	43.5
28	—	—	—	TAF 283720	—	—	51.5
	—	—	—	TAF 283730	—	—	83.5
	RNA 49/22	—	—	—	—	—	56.5
	—	RNA 69/22	—	—	—	—	97.5
29	—	—	—	TAF 293820	—	—	57
	—	—	—	TAF 293830	—	—	85
30	—	—	—	TAF 304020	—	—	64.5
	—	—	—	TAF 304030	—	—	97.5
	RNA 4905	—	—	—	—	—	64
	—	RNA 6905	—	—	—	—	111
	—	—	—	—	TR 304425	—	115
	—	—	—	—	—	GTR 304425	133

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
24	32	16	0.3	30	15 300	22 500	17 000
24	32	20	0.3	30	19 400	30 500	17 000
25	33	16	0.3	31	15 800	23 700	16 000
25	33	20	0.3	31	20 000	32 100	16 000
25	37	17	0.3	35	21 000	25 000	16 000
25	37	30	0.3	35	35 400	48 900	16 000
25	38	20	0.3	36	28 900	35 000	16 000
25	38	20	0.3	36	28 900	35 000	16 000
25	38	25	0.3	36	34 800	44 400	16 000
25	38	20	0.3	36	33 300	46 500	6 000
25	38	25	0.3	36	42 400	63 700	6 000
26	34	16	0.3	32	16 300	24 900	15 000
26	34	20	0.3	32	20 600	33 800	15 000
28	37	20	0.3	35	21 700	37 100	14 000
28	37	30	0.3	35	31 100	58 900	14 000
28	39	17	0.3	37	21 400	28 900	14 000
28	39	30	0.3	37	36 300	56 900	14 000
29	38	20	0.3	36	21 600	37 200	14 000
29	38	30	0.3	36	30 900	59 100	14 000
30	40	20	0.3	38	25 100	40 100	13 000
30	40	30	0.3	38	36 000	63 900	13 000
30	42	17	0.3	40	23 700	30 700	13 000
30	42	30	0.3	40	42 100	64 300	13 000
30	44	25	0.3	42	37 900	52 100	13 000
30	44	25	0.3	42	47 000	76 500	5 000

Notes(1) Minimum allowable value of chamfer dimension r

(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. TAF series with a roller set bore diameter F_w of 26 mm or less have no oil hole. In others, the outer ring has an oil groove and an oil hole.

2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring

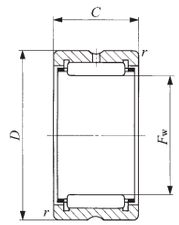




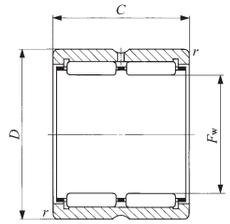
Shaft dia. 32 — 40mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
32	—	—	—	**TAF 324220**	—	—	68
	—	—	—	**TAF 324230**	—	—	102
	RNA 49/28	—	—	—	—	—	76.5
	—	**RNA 69/28**	—	—	—	—	133
	—	—	—	—	—	**GTR 324530**	152
35	—	—	—	**TAF 354520**	—	—	73.5
	—	—	—	**TAF 354530**	—	—	112
	RNA 4906	—	—	—	—	—	72.5
	—	**RNA 6906**	—	—	—	—	125
	—	—	—	—	**TR 354830**	—	139
	—	—	—	—	—	**GTR 354830**	163
37	—	—	—	**TAF 374720**	—	—	77.5
	—	—	—	**TAF 374730**	—	—	117
38	—	—	—	**TAF 384820**	—	—	79
	—	—	—	**TAF 384830**	—	—	119
	—	—	—	—	**TR 385230**	—	168
	—	—	—	—	—	**GTR 385230**	195
40	—	—	—	**TAF 405020**	—	—	83
	—	—	—	**TAF 405030**	—	—	125
	RNA 49/32	—	—	—	—	—	96
	—	**RNA 69/32**	—	—	—	—	172
	—	—	—	—	**TR 405520**	—	129
	—	—	—	—	—	**GTR 405520**	144

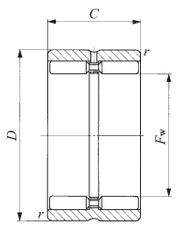
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.



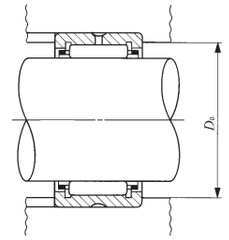
RNA49 TAF TR
RNA69 ($F_w \leqq 35$)



RNA69



GTR



D

NA
TAFI
TRI
BRI

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
32	42	20	0.3	40	25 700	42 200	12 000
32	42	30	0.3	40	36 800	67 200	12 000
32	45	17	0.3	43	24 500	32 700	12 000
32	45	30	0.3	43	41 800	64 800	12 000
32	45	30	0.3	43	58 000	101 000	4 500
35	45	20	0.3	43	26 900	46 200	11 000
35	45	30	0.3	43	38 600	73 600	11 000
35	47	17	0.3	45	25 200	34 700	11 000
35	47	30	0.3	45	43 000	69 000	11 000
35	48	30	0.3	46	47 400	72 300	11 000
35	48	30	0.3	46	61 100	110 000	4 500
37	47	20	0.3	45	28 200	50 100	11 000
37	47	30	0.3	45	40 500	79 800	11 000
38	48	20	0.3	46	28 100	50 200	11 000
38	48	30	0.3	46	40 300	80 000	11 000
38	52	30	0.6	48	50 800	81 100	11 000
38	52	30	0.6	48	64 200	121 000	4 000
40	50	20	0.3	48	29 400	54 100	10 000
40	50	30	0.3	48	42 300	86 200	10 000
40	52	20	0.6	48	31 200	47 800	10 000
40	52	36	0.6	48	53 500	95 700	10 000
40	55	20	0.6	51	37 400	55 700	10 000
40	55	20	0.6	51	44 300	73 600	3 500

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

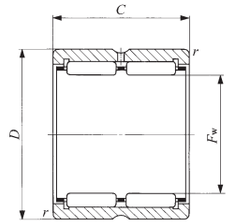
MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring

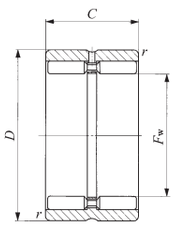




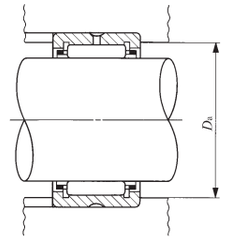
RNA49 TAF TR



RNA69



GTR



D

NA
TAFI
TRI
BRI

Shaft dia. 42 — 50mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
42	—	—	—	**TAF 425220**	—	—	86.5
	—	—	—	**TAF 425230**	—	—	130
	RNA 4907	—	—	—	—	—	113
	—	**RNA 6907**	—	—	—	—	200
	—	—	—	—	**TR 425630**	—	183
	—	—	—	—	—	**GTR 425630**	210
43	—	—	—	**TAF 435320**	—	—	88.5
	—	—	—	**TAF 435330**	—	—	133
45	—	—	—	**TAF 455520**	—	—	92
	—	—	—	**TAF 455530**	—	—	138
	RNA 49/38	—	—	—	—	—	120
	—	—	—	—	**TR 455930**	—	193
	—	—	—	—	—	**GTR 455930**	225
47	—	—	—	**TAF 475720**	—	—	95
	—	—	—	**TAF 475730**	—	—	144
48	**RNA 4908**	—	—	—	—	—	152
	—	—	—	—	**TR 486230**	—	205
	—	**RNA 6908**	—	—	—	—	275
	—	—	—	—	—	**GTR 486230**	240
50	—	—	—	**TAF 506225**	—	—	159
	—	—	—	**TAF 506235**	—	—	225
	—	—	—	—	**TR 506430**	—	210
	RNA 49/42	—	—	—	—	—	174
	—	—	—	—	—	**GTR 506430**	245

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
42	52	20	0.3	50	29 900	56 200	9 500
42	52	30	0.3	50	43 000	89 400	9 500
42	55	20	0.6	51	32 000	50 100	9 500
42	55	36	0.6	51	54 900	100 000	9 500
42	56	30	0.6	52	53 800	90 100	9 500
42	56	30	0.6	52	67 500	133 000	3 500
43	53	20	0.3	51	30 500	58 200	9 500
43	53	30	0.3	51	43 800	92 600	9 500
45	55	20	0.3	53	31 000	60 200	9 000
45	55	30	0.3	53	44 600	95 800	9 000
45	58	20	0.6	54	33 600	54 600	9 000
45	59	30	0.6	55	55 100	94 800	9 000
45	59	30	0.6	55	70 300	142 000	3 500
47	57	20	0.3	55	31 500	62 200	8 500
47	57	30	0.3	55	45 200	99 100	8 500
48	62	22	0.6	58	41 600	67 400	8 500
48	62	30	0.6	58	56 300	99 500	8 500
48	62	40	0.6	58	71 300	135 000	8 500
48	62	30	0.6	58	72 700	154 000	3 000
50	62	25	0.3	60	43 000	85 300	8 000
50	62	35	0.3	60	58 000	125 000	8 000
50	64	30	0.6	60	57 700	104 000	8 000
50	65	22	0.6	61	42 500	70 300	8 000
50	64	30	0.6	60	74 600	158 000	3 000

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring



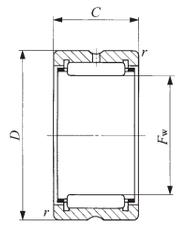




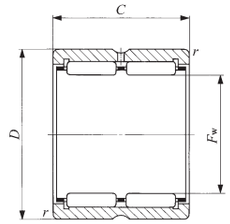
Shaft dia. 52 — 68mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
52	**RNA 4909**	—	—	—	—	—	197
	—	**RNA 6909**	—	—	—	—	355
55	—	—	—	**TAF 556825**	—	—	193
	—	—	—	**TAF 556835**	—	—	255
	RNA 49/48	—	—	—	—	—	188
58	**RNA 4910**	—	—	—	—	—	179
	—	**RNA 6910**	—	—	—	—	320
	—	—	—	—	**TR 587745**	—	515
	—	—	—	—	—	**GTR 587745**	590
60	—	—	—	**TAF 607225**	—	—	187
	—	—	—	**TAF 607235**	—	—	260
	RNA 49/52	—	—	—	—	—	205
62	—	—	—	—	**TR 628138**	—	460
	—	—	—	—	—	**GTR 628138**	520
63	**RNA 4911**	—	—	—	—	—	265
	—	**RNA 6911**	—	—	—	—	475
65	—	—	—	**TAF 657825**	—	—	225
	—	—	—	**TAF 657835**	—	—	315
	RNA 49/58	—	—	—	—	—	275
68	—	—	—	**TAF 688225**	—	—	250
	—	—	—	**TAF 688235**	—	—	350
	RNA 4912	—	—	—	—	—	285
	—	**RNA 6912**	—	—	—	—	510

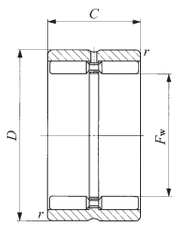
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.



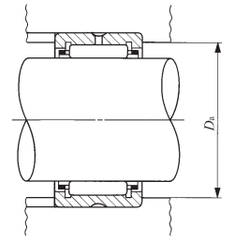
RNA49 TAF TR



RNA69



GTR



D

NA
TAFI
TRI
BRI

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
52	68	22	0.6	64	43 500	73 300	7 500
52	68	40	0.6	64	74 600	147 000	7 500
55	68	25	0.3	66	45 400	94 000	7 500
55	68	35	0.3	66	61 200	138 000	7 500
55	70	22	0.6	66	44 300	76 300	7 500
58	72	22	0.6	68	46 200	82 100	7 000
58	72	40	0.6	68	79 200	164 000	7 000
58	77	45	1	72	104 000	191 000	7 000
58	77	45	1	72	135 000	280 000	2 500
60	72	25	0.3	70	47 500	103 000	6 500
60	72	35	0.3	70	64 100	151 000	6 500
60	75	22	0.6	71	47 100	85 100	6 500
62	81	38	1	76	92 000	166 000	6 500
62	81	38	1	76	118 000	241 000	2 500
63	80	25	1	75	57 600	97 200	6 500
63	80	45	1	75	98 700	194 000	6 500
65	78	25	0.6	74	49 600	112 000	6 000
65	78	35	0.6	74	67 000	164 000	6 000
65	82	25	1	77	58 900	101 000	6 000
68	82	25	0.6	78	54 800	117 000	6 000
68	82	35	0.6	78	72 000	166 000	6 000
68	85	25	1	80	60 200	105 000	6 000
68	85	45	1	80	103 000	211 000	6 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring

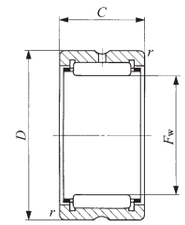




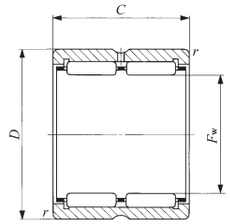
Shaft dia. 70 — 85mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
70	—	—	—	**TAF 708525**	—	—	280
	—	—	—	**TAF 708535**	—	—	395
	RNA 49/62	—	—	—	—	—	320
	—	—	—	—	**TR 708945**	—	605
	—	—	—	—	—	**GTR 708945**	690
72	**RNA 4913**	—	—	—	—	—	325
	—	**RNA 6913**	—	—	—	—	585
73	—	—	—	**TAF 739025**	—	—	335
	—	—	—	**TAF 739035**	—	—	475
75	—	—	—	**TAF 759225**	—	—	345
	—	—	—	**TAF 759235**	—	—	485
	RNA 49/68	—	—	—	—	—	470
80	—	—	—	**TAF 809525**	—	—	315
	—	—	—	**TAF 809535**	—	—	445
	RNA 4914	—	—	—	—	—	495
	—	**RNA 6914**	—	—	—	—	910
83	—	—	—	—	**TR 8310845**	—	995
	—	—	—	—	—	**GTR 8310845**	1 090
85	—	—	—	**TAF 8510525**	—	—	435
	RNA 4915	—	—	—	—	—	525
	—	—	—	**TAF 8510535**	—	—	610
	—	**RNA 6915**	—	—	—	—	960

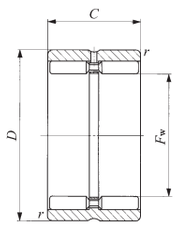
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.



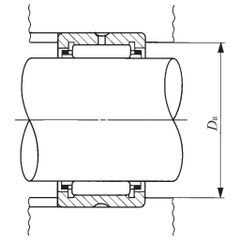
RNA49 TAF TR



RNA69



GTR



Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
70	85	25	0.6	81	55 500	120 000	5 500
70	85	35	0.6	81	73 000	171 000	5 500
70	88	25	1	83	61 500	109 000	5 500
70	89	45	1	84	114 000	228 000	5 500
70	89	45	1	84	147 000	336 000	2 000
72	90	25	1	85	62 700	113 000	5 500
72	90	45	1	85	108 000	227 000	5 500
73	90	25	1	85	61 100	127 000	5 500
73	90	35	1	85	80 400	181 000	5 500
75	92	25	1	87	62 100	131 000	5 500
75	92	35	1	87	81 700	186 000	5 500
75	95	30	1	90	79 900	147 000	5 500
80	95	25	1	90	59 400	137 000	5 000
80	95	35	1	90	78 100	195 000	5 000
80	100	30	1	95	83 200	158 000	5 000
80	100	54	1	95	134 000	311 000	5 000
83	108	45	1	103	146 000	270 000	5 000
83	108	45	1	103	190 000	396 000	1 800
85	105	25	1	100	76 300	145 000	4 500
85	105	30	1	100	86 200	169 000	4 500
85	105	35	1	100	102 000	210 000	4 500
85	105	54	1	100	138 000	331 000	4 500

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

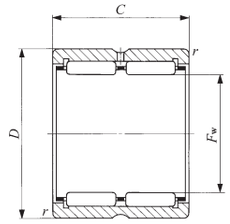
Without Inner Ring



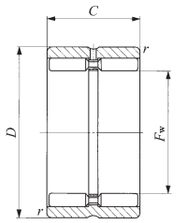




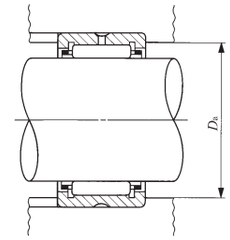
RNA49 TAF TR



RNA69



GTR



Shaft dia. 90 — 105mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
90	—	—	—	**TAF 9011025**	—	—	455
	RNA 4916	—	—	—	—	—	550
	—	—	—	**TAF 9011035**	—	—	640
	—	**RNA 6916**	—	—	—	—	1 010
93	—	—	—	—	**TR 9311850**	—	1 210
	—	—	—	—	—	**GTR 9311850**	1 340
95	—	—	—	**TAF 9511526**	—	—	495
	RNA 49/82	—	—	—	—	—	575
	—	—	—	**TAF 9511536**	—	—	690
	—	—	—	—	**TR 9512045**	—	1 120
	—	—	—	—	—	**GTR 9512045**	1 230
100	—	—	—	**TAF 10012026**	—	—	525
	RNA 4917	—	—	—	—	—	705
	—	—	—	**TAF 10012036**	—	—	725
	—	**RNA 6917**	—	—	—	—	1 300
	—	—	—	—	**TR 10012550**	—	1 290
	—	—	—	—	—	**GTR 10012550**	1 440
105	—	—	—	**TAF 10512526**	—	—	545
	RNA 4918	—	—	—	—	—	740
	—	—	—	**TAF 10512536**	—	—	760
	—	**RNA 6918**	—	—	—	—	1 360

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
90	110	25	1	105	77 300	150 000	4 500
90	110	30	1	105	87 300	175 000	4 500
90	110	35	1	105	103 000	217 000	4 500
90	110	54	1	105	143 000	351 000	4 500
93	118	50	1	113	165 000	329 000	4 500
93	118	50	1	113	224 000	509 000	1 600
95	115	26	1	110	79 700	159 000	4 000
95	115	30	1	110	90 000	186 000	4 000
95	115	36	1	110	106 000	231 000	4 000
95	120	45	1.5	112	155 000	305 000	4 000
95	120	45	1.5	112	204 000	455 000	1 600
100	120	26	1	115	82 400	168 000	4 000
100	120	35	1.1	113.5	110 000	244 000	4 000
100	120	36	1	115	110 000	244 000	4 000
100	120	63	1.1	113.5	173 000	467 000	4 000
100	125	50	1.5	117	172 000	355 000	4 000
100	125	50	1.5	117	234 000	549 000	1 500
105	125	26	1	120	84 700	178 000	4 000
105	125	35	1.1	118.5	113 000	258 000	4 000
105	125	36	1	120	113 000	258 000	4 000
105	125	63	1.1	118.5	178 000	490 000	4 000

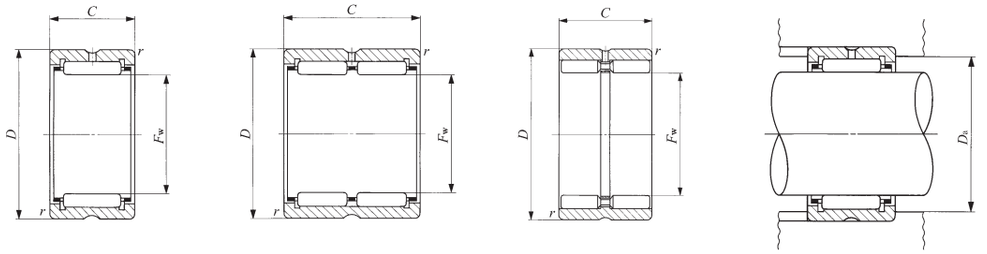
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring





RNA49 RNA48
TAF TR

RNA69

GTR

Shaft dia. 110 — 170mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
110	—	—	—	**TAF 11013030**	—	—	660
	RNA 4919	—	—	—	—	—	770
	—	—	—	**TAF 11013040**	—	—	880
	—	**RNA 6919**	—	—	—	—	1 420
	—	—	—	—	**TR 11013550**	—	1 400
	—	—	—	—	—	**GTR 11013550**	1 560
115	**RNA 4920**	—	—	—	—	—	1 190
	—	—	—	—	**TR 11515350**	—	2 350
	—	—	—	—	—	**GTR 11515350**	2 600
120	—	—	**RNA 4822**	—	—	—	790
125	**RNA 4922**	—	—	—	—	—	1 280
130	—	—	**RNA 4824**	—	—	—	850
135	**RNA 4924**	—	—	—	—	—	1 930
140	—	—	—	—	**TR 14017860**	—	3 320
	—	—	—	—	—	**GTR 14017860**	3 730
145	—	—	**RNA 4826**	—	—	—	1 100
150	**RNA 4926**	—	—	—	—	—	2 360
	—	—	—	—	**TR 15018860**	—	3 540
	—	—	—	—	—	**GTR 15018860**	3 970
155	—	—	**RNA 4828**	—	—	—	1 170
160	**RNA 4928**	—	—	—	—	—	2 500
165	—	—	**RNA 4830**	—	—	—	1 750
170	**RNA 4930**	—	—	—	—	—	4 090

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
110	130	30	1	125	106 000	240 000	3 500
110	130	35	1.1	123.5	116 000	271 000	3 500
110	130	40	1	125	134 000	324 000	3 500
110	130	63	1.1	123.5	182 000	514 000	3 500
110	135	50	1.5	127	183 000	395 000	3 500
110	135	50	1.5	127	245 000	603 000	1 400
115	140	40	1.1	133.5	145 000	329 000	3 500
115	140	40	1.1	133.5	145 000	329 000	3 500
115	153	50	1.5	145	233 000	414 000	3 500
115	153	50	1.5	145	315 000	614 000	1 300
120	140	30	1	135	93 200	239 000	3 500
125	150	40	1.1	143.5	152 000	357 000	3 000
130	150	30	1	145	96 900	259 000	3 000
135	165	45	1.1	158.5	187 000	435 000	3 000
140	178	60	1.5	170	307 000	625 000	3 000
140	178	60	1.5	170	409 000	923 000	1 100
145	165	35	1.1	158.5	116 000	340 000	3 000
150	180	50	1.5	172	215 000	540 000	2 500
150	188	60	1.5	180	320 000	675 000	2 500
150	188	60	1.5	180	423 000	989 000	1 000
155	175	35	1.1	168.5	120 000	363 000	2 500
160	190	50	1.5	182	224 000	580 000	2 500
165	190	40	1.1	183.5	168 000	446 000	2 500
170	210	60	2	201	324 000	712 000	2 500

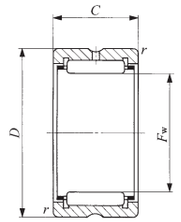
Notes(1) Minimum allowable value of chamfer dimension *r*
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Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

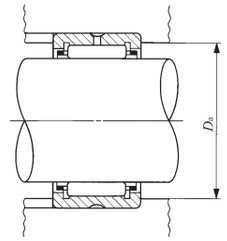
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring







RNA49 RNA48

Shaft dia. 175 — 350mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g
175	—	—	RNA 4832	—	—	—	1 850
180	RNA 4932	—	—	—	—	—	4 310
185	—	—	RNA 4834	—	—	—	2 700
190	RNA 4934	—	—	—	—	—	4 530
195	—	—	RNA 4836	—	—	—	2 840
205	RNA 4936	—	—	—	—	—	6 250
210	—	—	RNA 4838	—	—	—	3 380
215	RNA 4938	—	—	—	—	—	6 500
220	—	—	RNA 4840	—	—	—	3 520
225	RNA 4940	—	—	—	—	—	10 400
240	—	—	RNA 4844	—	—	—	3 820
245	RNA 4944	—	—	—	—	—	11 200
265	— RNA 4948	— —	RNA 4848	— —	— —	— —	5 670 12 000
285	—	—	RNA 4852	—	—	—	6 070
290	RNA 4952	—	—	—	—	—	21 200
305	—	—	RNA 4856	—	—	—	9 750
310	RNA 4956	—	—	—	—	—	22 500
330	—	—	RNA 4860	—	—	—	13 200
340	RNA 4960	—	—	—	—	—	33 400
350	—	—	RNA 4864	—	—	—	14 100

Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
175	200	40	1.1	193.5	173 000	474 000	2 500
180	220	60	2	211	337 000	761 000	1 900
185	215	45	1.1	208.5	211 000	567 000	1 900
190	230	60	2	221	347 000	810 000	1 900
195	225	45	1.1	218.5	218 000	602 000	1 900
205	250	69	2	241	434 000	989 000	1 900
210	240	50	1.5	232	249 000	726 000	1 800
215	260	69	2	251	440 000	1 020 000	1 700
220	250	50	1.5	242	255 000	766 000	1 600
225	280	80	2.1	269	518 000	1 120 000	1 600
240	270	50	1.5	262	266 000	833 000	1 500
245	300	80	2.1	289	536 000	1 200 000	1 400
265	300	60	2	291	345 000	1 150 000	1 300
265	320	80	2.1	309	565 000	1 320 000	1 300
285	320	60	2	311	354 000	1 220 000	1 100
290	360	100	2.1	349	847 000	1 900 000	1 100
305	350	69	2	341	486 000	1 550 000	950
310	380	100	2.1	369	877 000	2 040 000	950
330	380	80	2.1	369	610 000	1 900 000	900
340	420	118	3	407	1 130 000	2 650 000	850
350	400	80	2.1	389	635 000	2 040 000	750

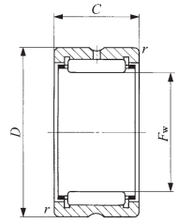
Notes(1) Minimum allowable value of chamfer dimension *r*
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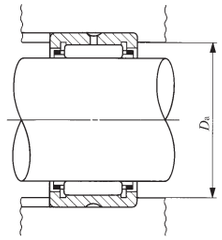
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1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring







RNA49 RNA48

Shaft dia. 360 — 490mm

Shaft dia. mm	Identification number RNA 49	RNA 69	RNA 48	TAF	TR	GTR	Mass (Ref.) g	Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
360	**RNA 4964**	—	—	—	—	—	35 200	360	440	118	3	427	1 170 000	2 830 000	750
370	—	—	**RNA 4868**	—	—	—	14 800	370	420	80	2.1	409	651 000	2 140 000	700
380	**RNA 4968**	—	—	—	—	—	37 000	380	460	118	3	447	1 220 000	3 020 000	700
390	—	—	**RNA 4872**	—	—	—	15 600	390	440	80	2.1	429	680 000	2 320 000	650
400	**RNA 4972**	—	—	—	—	—	38 700	400	480	118	3	467	1 260 000	3 200 000	600
415	—	—	**RNA 4876**	—	—	—	27 900	415	480	100	2.1	469	951 000	2 860 000	600
430	**RNA 4976**	—	—	—	—	—	56 400	430	520	140	4	504	1 540 000	4 030 000	500
450	**RNA 4980**	—	—	—	—	—	58 800	450	540	140	4	524	1 590 000	4 270 000	500
470	**RNA 4984**	—	—	—	—	—	61 200	470	560	140	4	544	1 640 000	4 510 000	500
490	**RNA 4988**	—	—	—	—	—	86 900	490	600	160	4	584	1 910 000	5 140 000	400

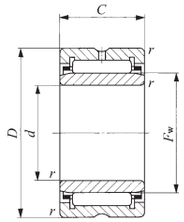
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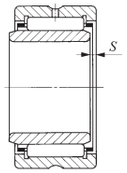
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1mm=0.03937inch

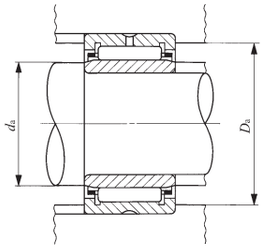
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring









NA49 TAFI
NA69 ($d \leqq 30$)

Shaft dia. 5 — 12mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
5	**NA 495**	—	—	—	—	—	7.3	5
	—	—	—	**TAFI 51512**	—	—	11.9	5
	—	—	—	**TAFI 51516**	—	—	16.7	5
6	**NA 496**	—	—	—	—	—	9.1	6
	—	—	—	**TAFI 61612**	—	—	13	6
	—	—	—	**TAFI 61616**	—	—	17.5	6
7	**NA 497**	—	—	—	—	—	11.2	7
	—	—	—	**TAFI 71712**	—	—	14.3	7
	—	—	—	**TAFI 71716**	—	—	19.2	7
8	**NA 498**	—	—	—	—	—	15	8
9	—	—	—	**TAFI 91912**	—	—	16.7	9
	—	—	—	**TAFI 91916**	—	—	22.5	9
	NA 499	—	—	—	—	—	16.7	9
10	**NA 4900**	—	—	—	—	—	24	10
	—	—	—	**TAFI 102216**	—	—	30	10
	—	—	—	**TAFI 102220**	—	—	38	10
12	**NA 4901**	—	—	—	—	—	26.5	12
	—	—	—	**TAFI 122416**	—	—	33.5	12
	—	—	—	**TAFI 122420**	—	—	42.5	12
	—	**NA 6901**	—	—	—	—	44.5	12

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
13	10	—	0.15	7	0.5	6.2	6.7	11.8	2 960	2 690	60 000	**LRT 5710**
15	12	—	0.2	8	0.5	6.6	7.7	13.4	5 060	4 690	50 000	**LRT 5812**
15	16	—	0.2	8	0.5	6.6	7.7	13.4	7 080	7 220	50 000	**LRT 5816**
15	10	—	0.15	8	0.5	7.2	7.7	13.8	3 960	3 420	50 000	**LRT 6810**
16	12	—	0.2	9	0.5	7.6	8.7	14.4	5 490	5 330	45 000	**LRT 6912**
16	16	—	0.2	9	0.5	7.6	8.7	14.4	7 680	8 210	45 000	**LRT 6916**
17	10	—	0.15	9	0.5	8.2	8.7	15.8	4 530	3 650	45 000	**LRT 7910**
17	12	—	0.2	10	0.5	8.6	9.7	15.4	5 880	5 970	40 000	**LRT 71012**
17	16	—	0.2	10	0.5	8.6	9.7	15.4	8 230	9 190	40 000	**LRT 71016**
19	11	—	0.2	10	0.5	9.6	9.9	17.4	6 180	5 030	40 000	**LRT 81011**
19	12	—	0.3	12	0.5	11	11.5	17	6 610	7 260	35 000	**LRT 91212**
19	16	—	0.3	12	0.5	11	11.5	17	9 250	11 200	35 000	**LRT 91216**
20	11	—	0.3	12	0.5	11	11.5	18	6 600	6 310	35 000	**LRT 91211**
22	13	—	0.3	14	0.5	12	13	20	9 230	10 100	30 000	**LRT 101413**
22	16	—	0.3	14	0.5	12	13	20	11 700	13 700	30 000	**LRT 101416**
22	20	—	0.3	14	0.5	12	13	20	14 800	18 600	30 000	**LRT 101420**
24	13	—	0.3	16	0.5	14	15	22	9 660	11 100	25 000	**LRT 121613**
24	16	—	0.3	16	0.5	14	15	22	12 300	15 100	25 000	**LRT 121616**
24	20	—	0.3	16	0.5	14	15	22	15 500	20 400	25 000	**LRT 121620**
24	22	—	0.3	16	0.5	14	15	22	17 100	23 000	25 000	**LRT 121622**

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. TAFI series with a bore diameter d of 22 mm or less have no oil hole. In others, the outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

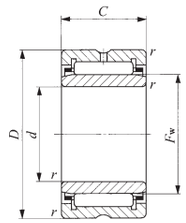
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

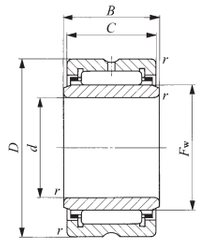
With Inner Ring



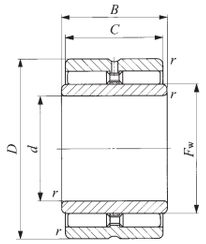




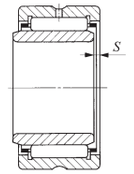
NA49 TAFI
NA69 ($d \leqq 30$)

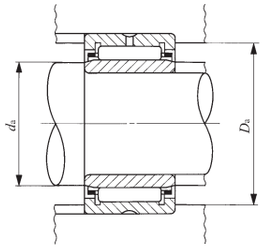


TRI



GTRI





Shaft dia. 15 — 22mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
15	—	—	—	**TAFI 152716**	—	—	39.5	15
	—	—	—	**TAFI 152720**	—	—	50	15
	NA 4902	—	—	—	—	—	35	15
	—	**NA 6902**	—	—	—	—	61	15
	—	—	—	—	**TRI 153320**	—	81	15
	—	—	—	—	—	**GTRI 153320**	90.5	15
17	—	—	—	**TAFI 172916**	—	—	43.5	17
	—	—	—	**TAFI 172920**	—	—	54	17
	NA 4903	—	—	—	—	—	39	17
	—	**NA 6903**	—	—	—	—	67	17
	—	—	—	—	**TRI 173425**	—	104	17
	—	—	—	—	—	**GTRI 173425**	117	17
20	—	—	—	**TAFI 203216**	—	—	48.5	20
	—	—	—	**TAFI 203220**	—	—	61	20
	NA 4904	—	—	—	—	—	78.5	20
	—	**NA 6904**	—	—	—	—	136	20
	—	—	—	—	**TRI 203820**	—	99	20
	—	—	—	—	**TRI 203825**	—	124	20
	—	—	—	—	—	**GTRI 203820**	110	20
	—	—	—	—	—	**GTRI 203825**	138	20
22	—	—	—	**TAFI 223416**	—	—	52	22
	—	—	—	**TAFI 223420**	—	—	67.5	22
	NA 49/22	—	—	—	—	—	87	22
	—	**NA 69/22**	—	—	—	—	152	22

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (3) rpm	Assembled inner ring
27	16	—	0.3	19	0.5	17	18	25	14 000	18 700	20 000	**LRT 151916**
27	20	—	0.3	19	0.5	17	18	25	17 700	25 300	20 000	**LRT 151920**
28	13	—	0.3	20	0.3	17	19	26	10 900	13 800	20 000	**LRT 152013**
28	23	—	0.3	20	0.3	17	19	26	19 300	28 800	20 000	**LRT 152023**
33	20	20.5	0.3	20	0.3	17	19	31	24 300	26 500	20 000	**LRT 152020**
33	20	20.5	0.3	20	—	17	19	31	29 200	37 200	7 500	**LRTZ 152020**
29	16	—	0.3	21	0.5	19	20	27	14 400	20 000	19 000	**LRT 172116**
29	20	—	0.3	21	0.5	19	20	27	18 200	27 100	19 000	**LRT 172120**
30	13	—	0.3	22	0.3	19	21	28	11 700	15 600	18 000	**LRT 172213**
30	23	—	0.3	22	0.3	19	21	28	20 800	32 500	18 000	**LRT 172223**
34	25	25.5	0.3	22	0.5	19	21	32	29 100	36 800	18 000	**LRT 172225**
34	25	25.5	0.3	22	—	19	21	32	37 900	57 800	7 000	**LRTZ 172225**
32	16	—	0.3	24	0.5	22	23	30	15 300	22 500	17 000	**LRT 202416**
32	20	—	0.3	24	0.5	22	23	30	19 400	30 500	17 000	**LRT 202420**
37	17	—	0.3	25	0.5	22	24	35	21 000	25 000	16 000	**LRT 202517**
37	30	—	0.3	25	0.5	22	24	35	35 400	48 900	16 000	**LRT 202530**
38	20	20.5	0.3	25	0.3	22	24	36	28 900	35 000	16 000	**LRT 202520**
38	25	25.5	0.3	25	0.5	22	24	36	34 800	44 400	16 000	**LRT 202525**
38	20	20.5	0.3	25	—	22	24	36	33 300	46 500	6 000	**LRTZ 202520**
38	25	25.5	0.3	25	—	22	24	36	42 400	63 700	6 000	**LRTZ 202525**
34	16	—	0.3	26	0.5	24	25	32	16 300	24 900	15 000	**LRT 222616**
34	20	—	0.3	26	0.5	24	25	32	20 600	33 800	15 000	**LRT 222620**
39	17	—	0.3	28	1	24	27	37	21 400	28 900	14 000	**LRT 222817**
39	30	—	0.3	28	0.5	24	27	37	36 300	56 900	14 000	**LRT 222830**

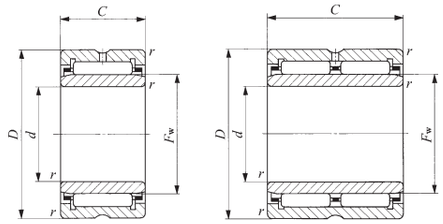
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. TAFI series with a bore diameter d of 22 mm or less have no oil hole. In others, the outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

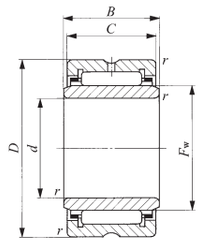
With Inner Ring



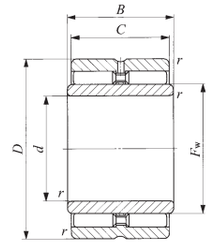


NA49 TAFI
NA69 ($d \leqq 30$)

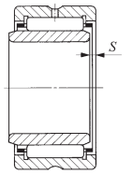
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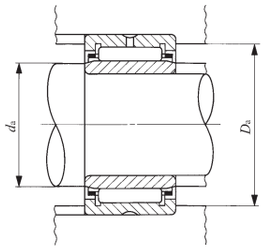


TRI



GTRI





Shaft dia. 25 — 32mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
25	—	—	—	**TAFI 253820**	—	—	82	25
	—	—	—	**TAFI 253830**	—	—	123	25
	NA 4905	—	—	—	—	—	92.5	25
	—	**NA 6905**	—	—	—	—	160	25
	—	—	—	—	**TRI 254425**	—	157	25
	—	—	—	—	—	**GTRI 254425**	175	25
28	—	—	—	**TAFI 284220**	—	—	96.5	28
	—	—	—	**TAFI 284230**	—	—	145	28
	NA 49/28	—	—	—	—	—	101	28
	—	**NA 69/28**	—	—	—	—	176	28
	—	—	—	—	—	**GTRI 284530**	196	28
30	—	—	—	**TAFI 304520**	—	—	112	30
	—	—	—	**TAFI 304530**	—	—	171	30
	NA 4906	—	—	—	—	—	106	30
	—	**NA 6906**	—	—	—	—	184	30
	—	—	—	—	**TRI 304830**	—	199	30
	—	—	—	—	—	**GTRI 304830**	225	30
32	—	—	—	**TAFI 324720**	—	—	121	32
	—	—	—	**TAFI 324730**	—	—	180	32
	NA 49/32	—	—	—	—	—	165	32
	—	—	—	—	**TRI 325230**	—	245	32
	—	**NA 69/32**	—	—	—	—	295	32
	—	—	—	—	—	**GTRI 325230**	270	32

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (3) rpm	Assembled inner ring
38	20	—	0.3	29	0.5	27	28	36	21 600	37 200	14 000	**LRT 252920**
38	30	—	0.3	29	1	27	28	36	30 900	59 100	14 000	**LRT 252930**
42	17	—	0.3	30	0.5	27	29	40	23 700	30 700	13 000	**LRT 253017**
42	30	—	0.3	30	0.5	27	29	40	42 100	64 300	13 000	**LRT 253030**
44	25	25.5	0.3	30	0.5	27	29	42	37 900	52 100	13 000	**LRT 253025**
44	25	25.5	0.3	30	—	27	29	42	47 000	76 500	5 000	**LRTZ 253025**
42	20	—	0.3	32	0.5	30	31	40	25 700	42 200	12 000	**LRT 283220**
42	30	—	0.3	32	1	30	31	40	36 800	67 200	12 000	**LRT 283230**
45	17	—	0.3	32	1	30	31	43	24 500	32 700	12 000	**LRT 283217**
45	30	—	0.3	32	1	30	31	43	41 800	64 800	12 000	**LRT 283230**
45	30	30.5	0.3	32	—	30	31	43	58 000	101 000	4 500	**LRTZ 283230**
45	20	—	0.3	35	0.3	32	34	43	26 900	46 200	11 000	**LRT 303520**
45	30	—	0.3	35	0.5	32	34	43	38 600	73 600	11 000	**LRT 303530**
47	17	—	0.3	35	0.5	32	34	45	25 200	34 700	11 000	**LRT 303517**
47	30	—	0.3	35	0.5	32	34	45	43 000	69 000	11 000	**LRT 303530**
48	30	30.5	0.3	35	1	32	34	46	47 400	72 300	11 000	**LRT 303530-1**
48	30	30.5	0.3	35	—	32	34	46	61 100	110 000	4 500	**LRTZ 303530**
47	20	—	0.3	37	0.3	34	36	45	28 200	50 100	11 000	**LRT 323720**
47	30	—	0.3	37	0.5	34	36	45	40 500	79 800	11 000	**LRT 323730**
52	20	—	0.6	40	0.5	36	39	48	31 200	47 800	10 000	**LRT 324020**
52	30	30.5	0.6	38	0.5	36	37	48	50 800	81 100	11 000	**LRT 323830**
52	36	—	0.6	40	0.3	36	39	48	53 500	95 700	10 000	**LRT 324036**
52	30	30.5	0.6	38	—	36	37	48	64 200	121 000	4 000	**LRTZ 323830**

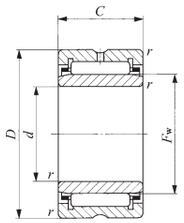
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

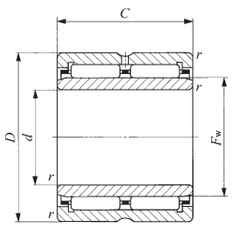
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring

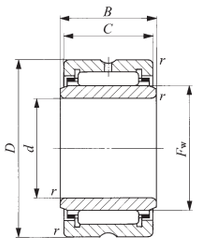




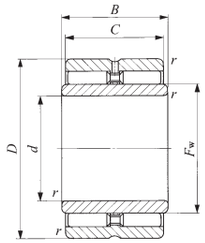
NA49 TAFI



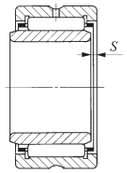
NA69

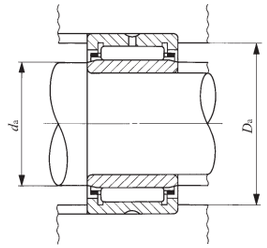


TRI



GTRI





Shaft dia. 35 — 45mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
35	—	—	—	**TAFI 355020**	—	—	129	35
	—	—	—	**TAFI 355030**	—	—	192	35
	NA 4907	—	—	—	—	—	178	35
	—	**NA 6907**	—	—	—	—	320	35
	—	—	—	—	**TRI 355630**	—	280	35
	—	—	—	—	—	**GTRI 355520**	191	35
	—	—	—	—	—	**GTRI 355630**	310	35
38	—	—	—	**TAFI 385320**	—	—	136	38
	—	—	—	**TAFI 385330**	—	—	205	38
40	—	—	—	**TAFI 405520**	—	—	143	40
	—	—	—	**TAFI 405530**	—	—	215	40
	—	—	—	—	**TRI 405930**	—	270	40
	NA 4908	—	—	—	—	—	245	40
	—	**NA 6908**	—	—	—	—	440	40
	—	—	—	—	—	**GTRI 405930**	300	40
42	—	—	—	**TAFI 425720**	—	—	149	42
	—	—	—	**TAFI 425730**	—	—	225	42
	—	—	—	—	**TRI 426230**	—	305	42
	—	—	—	—	—	**GTRI 426230**	340	42
45	—	—	—	**TAFI 456225**	—	—	230	45
	—	—	—	**TAFI 456235**	—	—	320	45
	—	—	—	—	**TRI 456430**	—	300	45
	NA 4909	—	—	—	—	—	285	45
	—	**NA 6909**	—	—	—	—	520	45
	—	—	—	—	—	**GTRI 456430**	335	45

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (3) rpm	Assembled inner ring
50	20	—	0.3	40	0.3	37	39	48	29 400	54 100	10 000	**LRT 354020**
50	30	—	0.3	40	0.5	37	39	48	42 300	86 200	10 000	**LRT 354030**
55	20	—	0.6	42	0.5	39	41	51	32 000	50 100	9 500	**LRT 354220**
55	36	—	0.6	42	0.3	39	41	51	54 900	100 000	9 500	**LRT 354236**
56	30	30.5	0.6	42	0.5	39	41	52	53 800	90 100	9 500	**LRT 354230**
55	20	20.5	0.6	40	—	39	39.5	51	44 300	73 600	3 500	**LRTZ 354020**
56	30	30.5	0.6	42	—	39	41	52	67 500	133 000	3 500	**LRTZ 354230**
53	20	—	0.3	43	0.3	40	42	51	30 500	58 200	9 500	**LRT 384320**
53	30	—	0.3	43	0.5	40	42	51	43 800	92 600	9 500	**LRT 384330**
55	20	—	0.3	45	0.3	42	44	53	31 000	60 200	9 000	**LRT 404520**
55	30	—	0.3	45	0.5	42	44	53	44 600	95 800	9 000	**LRT 404530**
59	30	30.5	0.6	45	1	44	44.5	55	55 100	94 800	9 000	**LRT 404530-1**
62	22	—	0.6	48	0.5	44	47	58	41 600	67 400	8 500	**LRT 404822**
62	40	—	0.6	48	0.3	44	47	58	71 300	135 000	8 500	**LRT 404840**
59	30	30.5	0.6	45	—	44	44.5	55	70 300	142 000	3 500	**LRTZ 404530**
57	20	—	0.3	47	0.3	44	46	55	31 500	62 200	8 500	**LRT 424720**
57	30	—	0.3	47	0.5	44	46	55	45 200	99 100	8 500	**LRT 424730**
62	30	30.5	0.6	48	0.5	46	47	58	56 300	99 500	8 500	**LRT 424830**
62	30	30.5	0.6	48	—	46	47	58	72 700	154 000	3 000	**LRTZ 424830**
62	25	—	0.3	50	0.5	47	49	60	43 000	85 300	8 000	**LRT 455025**
62	35	—	0.3	50	1	47	49	60	58 000	125 000	8 000	**LRT 455035**
64	30	30.5	0.6	50	1	49	49.5	60	57 700	104 000	8 000	**LRT 455030**
68	22	—	0.6	52	0.5	49	51	64	43 500	73 300	7 500	**LRT 455222**
68	40	—	0.6	52	0.3	49	51	64	74 600	147 000	7 500	**LRT 455240**
64	30	30.5	0.6	50	—	49	49.5	60	74 600	158 000	3 000	**LRTZ 455030**

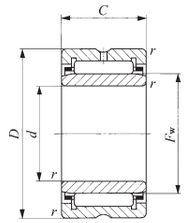
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

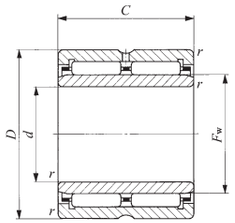
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring

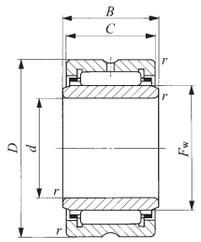




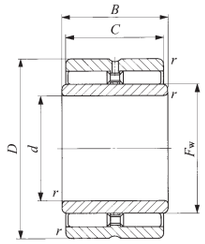
NA49 TAFI



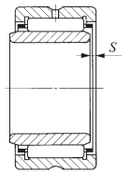
NA69

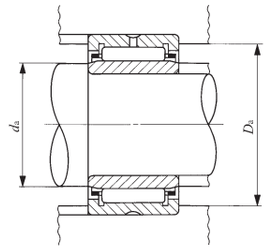


TRI



GTRI





Shaft dia. 50 — 70mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
50	—	—	—	**TAFI 506825**	—	—	270	50
	—	—	—	**TAFI 506835**	—	—	365	50
	NA 4910	—	—	—	—	—	295	50
	—	**NA 6910**	—	—	—	—	530	50
	—	—	—	—	**TRI 507745**	—	755	50
	—	—	—	—	—	**GTRI 507745**	825	50
55	—	—	—	**TAFI 557225**	—	—	275	55
	—	—	—	**TAFI 557235**	—	—	380	55
	NA 4911	—	—	—	—	—	410	55
	—	**NA 6911**	—	—	—	—	730	55
	—	—	—	—	**TRI 558138**	—	650	55
	—	—	—	—	—	**GTRI 558138**	710	55
60	—	—	—	**TAFI 608225**	—	—	395	60
	—	—	—	**TAFI 608235**	—	—	560	60
	NA 4912	—	—	—	—	—	440	60
	—	**NA 6912**	—	—	—	—	785	60
	—	—	—	—	**TRI 608945**	—	960	60
	—	—	—	—	—	**GTRI 608945**	1 050	60
65	**NA 4913**	—	—	—	—	—	470	65
	—	—	—	**TAFI 659035**	—	—	710	65
	—	**NA 6913**	—	—	—	—	840	65
70	—	—	—	**TAFI 709525**	—	—	540	70
	—	—	—	**TAFI 709535**	—	—	755	70
	NA 4914	—	—	—	—	—	765	70
	—	**NA 6914**	—	—	—	—	1 400	70

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
68	25	—	0.3	55	0.5	52	54	66	45 400	94 000	7 500	**LRT 505525**
68	35	—	0.3	55	1	52	54	66	61 200	138 000	7 500	**LRT 505535**
72	22	—	0.6	58	0.5	54	57	68	46 200	82 100	7 000	**LRT 505822**
72	40	—	0.6	58	0.3	54	57	68	79 200	164 000	7 000	**LRT 505840**
77	45	45.5	1	58	2	55	57	72	104 000	191 000	7 000	**LRT 505845**
77	45	45.5	1	58	—	55	57	72	135 000	280 000	2 500	**LRTZ 505845**
72	25	—	0.3	60	0.5	57	59	70	47 500	103 000	6 500	**LRT 556025**
72	35	—	0.3	60	1	57	59	70	64 100	151 000	6 500	**LRT 556035**
80	25	—	1	63	1	60	61	75	57 600	97 200	6 500	**LRT 556325**
80	45	—	1	63	0.5	60	61	75	98 700	194 000	6 500	**LRT 556345**
81	38	38.5	1	62	1.5	60	60.5	76	92 000	166 000	6 500	**LRT 556238**
81	38	38.5	1	62	—	60	60.5	76	118 000	241 000	2 500	**LRTZ 556238**
82	25	—	0.6	68	0.3	64	66	78	54 800	117 000	6 000	**LRT 606825**
82	35	—	0.6	68	1	64	66	78	72 000	166 000	6 000	**LRT 606835**
85	25	—	1	68	1	65	66	80	60 200	105 000	6 000	**LRT 606825-1**
85	45	—	1	68	0.5	65	66	80	103 000	211 000	6 000	**LRT 606845**
89	45	45.5	1	70	2	65	68	84	114 000	228 000	5 500	**LRT 607045**
89	45	45.5	1	70	—	65	68	84	147 000	336 000	2 000	**LRTZ 607045**
90	25	—	1	72	1	70	70.5	85	62 700	113 000	5 500	**LRT 657225**
90	35	—	1	73	1	70	71	85	80 400	181 000	5 500	**LRT 657335**
90	45	—	1	72	0.5	70	70.5	85	108 000	227 000	5 500	**LRT 657245**
95	25	—	1	80	0.3	75	78	90	59 400	137 000	5 000	**LRT 708025**
95	35	—	1	80	1	75	78	90	78 100	195 000	5 000	**LRT 708035**
100	30	—	1	80	1.5	75	78	95	83 200	158 000	5 000	**LRT 708030**
100	54	—	1	80	1	75	78	95	134 000	311 000	5 000	**LRT 708054**

Notes(1) Minimum allowable value of chamfer dimension r
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Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch